



Phase 1 SaaS Architecture

Limits · Costs · 500 Devices · 2026

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Platform planning · Version 1.0

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NYC1 Phase 1 · SaaS + MQTT + Archive + Managed Mongo + Cellular

Executive Summary

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mooreVIEW Phase 1 — SaaS Architecture, Limits & Cost Model

500 devices · 32 camera/appliance uplinks · NYC1 Phase 1 topology · 2026

Document version: 1.0 · 2026-08-13

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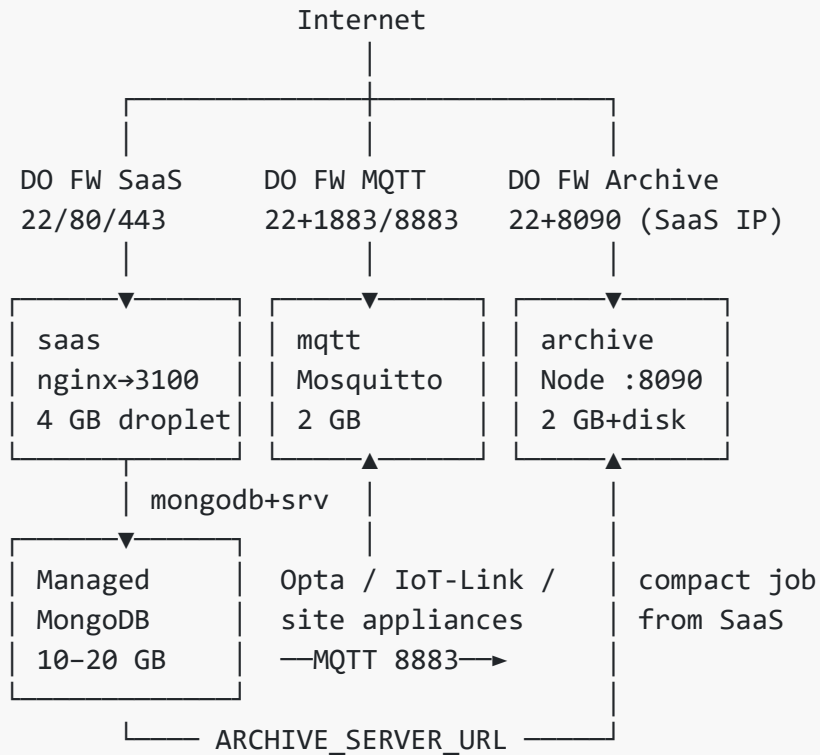
Executive summary

Phase 1 production topology for mooreVIEW Cloud SaaS: **three DigitalOcean droplets** (SaaS, dedicated MQTT, archive) plus **Managed MongoDB**. This document adds **device/tag capacity limits** and a **2026 cost model** for a fleet of **500 field-service devices** and **32 camera/appliance cellular uplinks**.

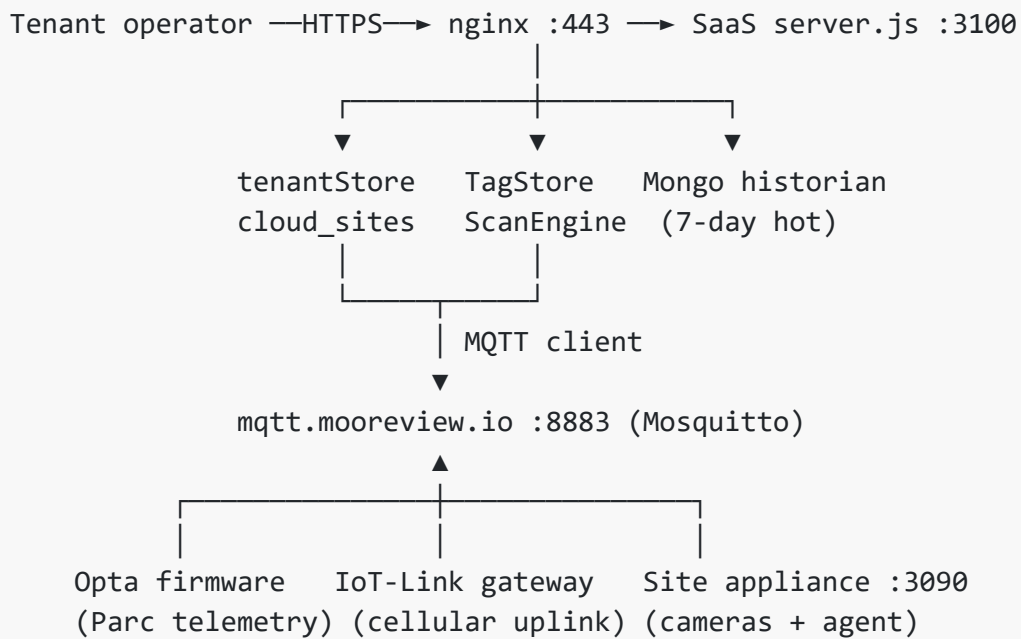
Category	Monthly	Annual
Gross renewal revenue (500 × \$10/mo)	\$5,000	\$60,000
DO Phase 1 cloud stack	~\$93	~\$1,119
Cellular — 500 field devices (\$2 SIM + \$0.50/50 MB)	\$1,250	\$15,000
Cellular — 32 cameras/appliances (\$2 SIM + \$5/1 GB)	\$224	\$2,688
Total infrastructure (cloud + cellular)	~\$1,567	~\$18,807
Contribution after infra	~\$3,433	~\$41,193
Infra as % of revenue	~31.3%	~31.3%

At 500 devices, cloud infrastructure alone remains **~1.9% of revenue**; cellular field connectivity dominates operating cost at **~29.5%**.

Phase 1 architecture



Data & control flow



SaaS application stack (:3100)

Layer	Components
Auth	/login — org + email; tenantStore; JWT/session; roles (operator, tenant_admin, platform_admin)
Cloud Studio	ST program, tags, HMI, drivers, alarms, historian — same runtime as appliance
Fleet	/sites pairing, site agent heartbeat, /fleet map, /sites/devices inventory
Integrations	CMMS entitlement, Cloud Sims, Cellular SIM vendor registry
Runtime core	TagStore → ScanEngine → DriverManager; eventBus → alarmNotifier
Data	cloud_tenants.json / Mongo; cloud_sites.json; /var/lib/mooreview workspace; Mongo historian

Phase 1 capacity limits (max)

Directional limits for a **single Phase 1 stack** before the next DO tier or VM split (see ARCHITECTURE.md).

Platform & tenant limits

Resource	Phase 1 max (configured)	Phase 1 recommended @ 500 devices	Notes
Devices per tenant	1,000	500	Configurable tenant quota; ARCHITECTURE.md
Locations / sites per tenant	1,000	~50–200	Sites = paired edge appliances
Users per tenant	10,000 (target)	5–50	Cloud target scale
Locations on one stack	~10–50 (directional)	1–5	Depends on aggregate device count
MQTT connections (brocr)			

500-device fleet (minimal)	500 × 3 pens + 2, +5% headroom	1,664
Appliance (edge PC)	Default	4,096

Override at deploy time: `MOOREVIEW_MAX_TAGS=3328` in `/etc/mooreview/saas.env` .

Device counts @ 500-device reference fleet

Device class	Count	Uplink	Data plan
Field-service monitors (Opta / IoT-Link)	500	Cellular MQTT	\$2/SIM + \$0.50 per 50 MB/mo
Camera / appliance uplinks (MVP Suite site agents)	32	Cellular	\$2/SIM + \$5 per 1 GB/mo
Total MQTT endpoints	532	—	—

Storage & ingest @ 500 + 32

Layer	Field fleet (500 × 50 MB/mo)	Camera/appliance (32 × ~100 MB/mo est.)	Combined
Monthly ingest	25 GB	~3.2 GB	~28 GB
Hot Mongo (7-day rolling)	~5.8 GB	~0.7 GB	~6.5 GB
Archive Year 1 cumulative	~300 GB	~38 GB	~338 GB

Hot tier (6.5 GB) fits within the ~~15 GB~~ base Managed Mongo plan. Archive volume: start ~~200 GB~~, add ~~**150 GB**~~ (~\$15/mo) before month 12.

When to scale beyond Phase 1

Signal	Threshold	Next step
Devices per tenant	> 1,000	Tenant shard or second SaaS node
Total fleet devices	> ~2,000 on one MQTT droplet	Dedicated ingest VM or broker cluster
Tag count	> 70,912 (or env cap)	Raise <code>MOOREVIEW_MAX_TAGS</code> ; scale Mongo compute — see <code>Phase1-Tag-Scaling-1M-5M-25M.md</code>
Hot Mongo	> 15 GB steady	Add scalable storage (\$0.215/GiB/mo)
Archive	> 500 GB	Expand block volume; consider object storage
SaaS CPU/RAM	Sustained >70% on 4 GB	Step to 8 GB droplet (\$48/mo)

Long-term target (`ARCHITECTURE.md`): **10M devices, 10K users** with sharded ingestion and VM split (ingestion · alarms · GUI · AI).

ALF large-tag fleets: 60 campuses × 4,000 on-prem tags — use **rollup site-agent sync** (~600 cloud tags/campus). Full mirror breaks default **70,912** tag cap at ~18 campuses. See **MooreVIEW-Infrastructure-Projections.md** Part 10.

Cloud infrastructure cost @ 500 devices

DigitalOcean list pricing (Aug 2026). See also **DO-Phase1-Storage-Cost-Review.md** .

Component	Monthly	Annual
SaaS 4 GB droplet	\$24	\$288
MQTT 2 GB droplet	\$12	\$144
Archive 2 GB droplet	\$12	\$144
Archive volume (~338 GB Y1 avg)	~\$34	~\$408
Managed Mongo (15 GB base)	\$15.23	\$183
Cloud subtotal	~\$97	~\$1,167

Per device (cloud only): ~~\$0.19/device/month~~ (**\$2.33/device/year**) — **1.9% of \$10/mo renewal**.

Cellular cost — field service fleet (500 devices)

Assumes each monitored asset uses one IoT SIM for MQTT Parc uplink.

Line	Rate	Monthly	Annual
SIM fee	\$2.00/device/mo	\$1,000	\$12,000
Data (50 MB/mo included plan)	\$0.50/device/mo	\$250	\$3,000
Field fleet subtotal	\$2.50/device/mo	\$1,250	\$15,000

Per device: \$2.50/mo (**25% of \$10/mo renewal**).

50 MB/mo is sufficient for sparse telemetry (5 s sample, compressed Parc JSON). High-rate PdM or firmware OTA may require higher data tiers.

Cellular cost — camera / appliance uplinks (32 units)

Site appliances (MVP Suite on **3090**) with cellular for site-agent heartbeat, inventory sync, and remote camera proxy.

Line	Rate	Monthly	Annual
SIM fee	\$2.00/unit/mo	\$64	\$768
Data (1 GB/mo plan)	\$5.00/unit/mo	\$160	\$1,920
Camera/appliance subtotal	\$7.00/unit/mo	\$224	\$2,688

Per unit: \$7.00/mo. These 32 units are typically **enterprise/campus** sites (ALF, multi-building) — often billed at higher SaaS tiers; shown here for infra planning only.

1 GB/mo supports site-agent heartbeat, inventory sync, and occasional camera thumbnail/live-view proxy. Continuous HD streaming requires a higher tier or local LAN-only cameras.

Combined cost & margin (2026 reference fleet)

Fleet: **500 field devices + 32 camera/appliance uplinks** · renewal **\$10/device/mo** on the 500 field assets.

	Monthly	Annual	% of \$5,000/mo revenue
Gross renewal revenue (500 × \$10)	\$5,000	\$60,000	100%
DO Phase 1 cloud stack	−\$97	−\$1,167	1.9%
Cellular — 500 field devices	−\$1,250	−\$15,000	25.0%
Cellular — 32 camera/appliance	−\$224	−\$2,688	4.5%
Total infrastructure	−\$1,571	−\$18,855	31.4%
Contribution after infra	\$3,429	\$41,145	68.6%

Per-device economics (500 field devices)

	Per device/mo	Per device/yr
Renewal revenue	\$10.00	\$120.00
Cloud infra (allocated)	~\$0.19	~\$2.33
Cellular (field)	\$2.50	\$30.00
Camera/appliance cell (allocated)	~\$0.45	~\$5.38
Contribution after infra	~\$6.86	~\$82.29

Camera/appliance cellular (\$224/mo) allocated across 500 renewal devices for portfolio view; actual billing may attach to enterprise site fees.

Break-even (infrastructure only)

Cost covered	Devices @ \$10/mo
Cloud stack alone (~\$97/mo)	~10 devices
Cloud + field cellular (~\$1,347/mo)	~135 devices
Full stack incl. 32 uplinks (~\$1,571/mo)	~157 devices

Revenue vs cost summary

Revenue (500 devices × \$10/mo)	\$5,000/mo		100%
Cloud (D0 Phase 1)	-\$97/mo		1.9%
Cellular – field (500 × \$2.50)	-\$1,250/mo		25.0%
Cellular – cameras (32 × \$7)	-\$224/mo		4.5%
Contribution	\$3,429/mo		68.6%

Reference configuration (production)

Host	Role	Size
cloud-1-saas-nyc1	SaaS nginx → :3100	4 GB / 2 vCPU / 80 GB SSD
cloud-mqtt-nyc	Mosquitto 1883/8883	2 GB / 1 vCPU / 50 GB SSD
archive-prod-nyc	Archive API :8090	2 GB / 1 vCPU + block volume
Managed MongoDB	Hot historian (7-day rolling)	10–20 GB tier

DNS: mooreview.io → SaaS · mqtt.mooreview.io → MQTT · archive.mooreview.io → Archive (optional)

Env: /etc/mooreview/saas.env — MONGODB_URI , JWT_SECRET ,
MOOREVIEW_MQTT_BROKER=mqtts://mqtt.mooreview.io:8883 , ARCHIVE_SERVER_URL ,
MOOREVIEW_MAX_TAGS

Assumptions & disclaimer

Assumption	Value
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Field devices	500 on \$10/mo renewal
Camera/appliance uplinks	32 (enterprise sites; not in the 500 renewal count)
Field cellular	\$2/SIM + \$0.50 per 50 MB/mo
Camera/appliance cellular	\$2/SIM + \$5 per 1 GB/mo
Field ingest	50 MB/device/month
Camera ingest (est.)	~100 MB/unit/month
Hot retention	7 days Mongo → zstd archive
Region	NYC1 / NYC3 (DigitalOcean)
Topology	Phase 1: SaaS + MQTT + Archive + Managed Mongo

Not included: hardware/commissioning, DNS, payment processing, support labor, sales/marketing, firmware, Opta/IoT-Link capex, overage charges beyond stated data plans.

Disclaimer: Directional planning document for architecture, capacity, and opex review. Not a financial forecast, hardware quote, or SLA commitment.

Related documents

Document	Purpose
CLOUD_DEPLOY_DO_PHASE1_ATL-MQTT.md	Phase 1 deploy runbook
DO-Phase1-Storage-Cost-Review.md	Storage-only cost deep dive
ARCHITECTURE.md	Appliance vs cloud roadmap
CLOUD_SAAS.md	SaaS modes, routes, seed accounts
MooreVIEW-Infrastructure-Projections.md	5-year portfolio scaling
Phase1-Tag-Scaling-1M-5M-25M.md	1M / 5M / 25M tags · DO elastic · when to switch

Phase 1 SaaS Architecture & Cost Model v1.0 — mooreVIEW platform planning.

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